

Physiological Signal Platform — Mathematical Reference

1. Physiological Signal Modeling

Effective Heart Rate:

$$HR_{eff} = HR_{base} + k_A \cdot A$$

Effective Respiration Rate:

$$RR_{eff} = RR_{base} + k_R \cdot A$$

ECG Component:

$$ECG(t) = A_{ecg} \sin(2\pi f_{heart} t)$$

Respiration Component:

$$RESP(t) = A_{resp} \sin(2\pi f_{resp} t)$$

EMG (Band-limited Noise Approximation):

$$EMG(t) = \mathcal{N}(0, \sigma^2)$$

Skin Conductance Drift:

$$GSR(t) = A_{gsr} \sin(2\pi f_{slow} t)$$

Where A represents global sympathetic activation.

2. Analog Front-End Conditioning

Differential Amplifier:

$$V_{amp} = G \cdot (V^+ - V^-)$$

First-Order Low-Pass Filter:

$$f_c = \frac{1}{2\pi RC}$$

Discrete-Time Filter Approximation:

$$y[n] = y[n - 1] + \alpha(x[n] - y[n - 1])$$

$$\alpha = \frac{dt}{RC + dt}$$

Saturation Clamp:

$$V_{out} = \begin{cases} V_{max}, & V > V_{max} \\ V_{min}, & V < V_{min} \\ V, & \text{otherwise} \end{cases}$$

3. Analog-to-Digital Conversion

Quantization:

$$ADC = \left\lfloor \frac{V_{out}}{V_{ref}} \cdot (2^N - 1) \right\rfloor$$

Where N is ADC resolution (e.g., 10-bit).

4. State Inference Projection

Valence Mapping:

$$Valence = \frac{ADC}{ADC_{max}} - 0.5$$

Energy Mapping:

$$Energy = A - 0.5$$

These are projected into a 2D affective state space.

5. Scenario Mode (Demonstration Presets)

Select a preset to simulate system-wide modulation:

Rest

Cognitive Stress

Physical Activity

Baseline resting physiology.